

📄 Key SQL Queries

Below are selected SQL queries used to analyze customer churn and identify key business drivers.

📊 1. Churn Distribution

```
``sql
SELECT
    churn,
    COUNT(*) AS customers
FROM churn
GROUP BY churn;
``
```



Purpose:

Provides a baseline view of how many customers stayed vs churned.

📈 2. Overall Churn Rate

```
``sql
SELECT
    ROUND(AVG(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) * 100, 2) AS
    churn_rate_percent
FROM churn;
``
```



Purpose:

Calculates the percentage of customers who churned.

📌 3. Churn by Contract Type

```
``sql
SELECT
    contract,
    COUNT(*) AS churned_customers
FROM churn
WHERE churn = 'Yes'
```

```
GROUP BY contract
ORDER BY churned_customers DESC;
'''
```

💡 Purpose:
Identifies which contract types contribute most to churn.

🌐 4. Churn by Internet Service

```
```sql
SELECT
 internet_service,
 COUNT(*) AS churned_customers
FROM churn
WHERE churn = 'Yes'
GROUP BY internet_service
ORDER BY churned_customers DESC;
'''
```

💡 Purpose:  
Analyzes how different service types impact churn.

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#### ### 🔗 5. High-Risk Customer Segments

```
```sql
SELECT
    contract,
    internet_service,
    COUNT(*) AS churned_customers
FROM churn
WHERE churn = 'Yes'
GROUP BY contract, internet_service
ORDER BY churned_customers DESC;
'''
```

💡 Purpose:
Identifies combinations of attributes associated with the highest churn.

🇵🇹 6. Churn by Payment Method (Counts)

```
```sql
SELECT
 payment_method,
 COUNT(*) AS churned_customers
FROM churn
WHERE churn = 'Yes'
GROUP BY payment_method
ORDER BY churned_customers DESC;
```
```



Purpose:

Highlights which payment methods are most common among churned customers.

🇺🇸 7. Churn Rate by Payment Method

```
```sql
SELECT
 payment_method,
 ROUND(AVG(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) * 100, 2) AS
 churn_rate_percent
FROM churn
GROUP BY payment_method
ORDER BY churn_rate_percent DESC;
```
```



Purpose:

Compares churn risk across payment methods using percentage-based analysis.
